Lab Manual: Practical 6

Course Code: ETCS406

Course Title: Game Development

Program: B. Tech Computer Science and Engineering

Semester: VII

Practical Title: Design and Develop a Narrative-Based Interactive Story Game Using Unity's Timeline and Animator

Duration: 3 Hours

Lab Type: Narrative Design & Animation

Assessment Weightage: Part of CAL 1 (Lab Assignments)

Objective

To design and implement a narrative-driven interactive story game using Unity's Timeline and Animator tools. Students will learn how to create character-driven storytelling experiences with animations, transitions, and user interaction.

Pre-requisites

- Basic Unity Editor navigation
- Familiarity with C# scripting
- Understanding of Unity Animator and Timeline components
- Exposure to storytelling techniques and branching narratives

Learning Outcomes

By the end of this lab, students will be able to:

- Create a linear or branching story structure for a game
- Use Unity's Timeline to sequence animations and events
- Implement character animations using Animator Controller
- Build interactive story elements with user choices and transitions

Tools & Resources

- Unity Editor (2020.3 LTS or later recommended)

- Visual Studio or any C# IDE
- Unity Timeline and Animator packages
- Free assets from Unity Asset Store (characters, backgrounds, audio)
- Reference: Unity Learn – Creating Cutscenes with Timeline

Lab Tasks

Task 1: Story Planning

- Define the story concept: setting, characters, conflict, resolution
- Decide on linear or branching structure
- Write a short script or storyboard for the game

Task 2: Scene Setup

- Create a new Unity 2D or 3D project
- Design the environment and import character assets
- Set up UI elements for dialogue and choices

Task 3: Timeline & Animator Integration

- Use **Timeline** to create cutscenes and event sequences
- Add **Animator Controller** to characters for movement and expressions
- Trigger animations and transitions based on story progression

Task 4: Interactivity

- Implement dialogue system using C# scripts
- Add player choices that affect story flow (e.g., branching paths)
- Use triggers and conditions to control scene transitions

Task 5: Testing & Polish

- Playtest the story game for flow and engagement
- Debug animation timing, transitions, and UI behavior
- Add sound effects or background music for immersion

Evaluation Criteria

Component	Marks
Story Design & Structure	3
Timeline & Animator Usage	3
Interactivity & Dialogue System	2
Code Quality & Debugging	2
Total	**10**

Tips for Success

- Keep the story concise but emotionally engaging
- Use Timeline for smooth transitions and pacing
- Test player choices thoroughly to avoid dead ends
- Use animations to enhance character personality and mood

Suggested Reading & Resources

- *The Art of Game Design* by Jesse Schell – Chapters on storytelling and player experience
- Unity Learn: Timeline and Cinematics
- Brackeys YouTube: Dialogue System Tutorial

Submission Guidelines:

- Submit your report in PDF format via the Google Classroom.
- Include screenshots or gameplay references where relevant.